Marine Policy 154 (2023) 105689

Contents lists available at ScienceDirect

Marine Policy

journal homepage: www.elsevier.com/locate/marpol

Framework system of marine sustainable development assessment based on systematic review

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ARTICLE INFO

Keywords:
Systematic review
Marine sustainable development assessment
Sustainability levels
Conceptual framework
Evaluation framework system

ABSTRACT

In the past two decades, the connotation, assessment method, assessment indicators, data source and monitoring means of marine sustainable development have been developing continuously. A systematic review of the assessment of marine sustainable development is helpful to clarify the characteristics and trends of this study and provide guidance for follow-up research. Taking marine sustainable development assessment as the key word, this study analyzes the conceptual framework and evaluation framework system of marine sustainable development by extensively searching several online databases and adopting the method of systematic review. According to the researchers' understanding of different levels of sustainable development and their relationships, the concept of marine sustainable development is summarized, and the conceptual framework of two-dimensional, three-dimensional, four-dimensional and five-dimensional levels is put forward and explained. Based on the framework of marine sustainable development assessment, including assessment objectives, assessment methods, assessment scope, data collection, analysis of assessment results and policy recommendations, this study summarizes previous studies, highlights the characteristics of each link in the process of marine sustainable development assessment, points out the limitations of contemporary research literature and problems that need to be solved in the future, and puts forward suggestions for future research.

1. Introduction

In 1987, the United Nations World Commission on Environment and Development published the Brundtland Report. Since then, the terms "sustainability" and "sustainable development" became very commonplace [1]. The former focused on the state [2,3], while the latter focused on the process[4]. Sustainable development was defined as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" [5–7]. This is recognized as the first definition of sustainable development in the world. Since then, different experts, scholars and institutions have expounded and deduced the concept and connotation of sustainable development from various aspects. To sum up, the main points are as follows:

(1) Pay attention to the relationship between man and nature. Human beings should adhere to a harmonious way with nature [7], maintain a healthy earth, bring necessary benefits to all people, pursue a healthy and productive life, and do not seek development in a way that exhausts resources, pollutes the environment and destroys ecology.

(2) Pay attention to the relationship among people. The fairness of development is reflected[8–12],including intra-generational fairness and intergenerational fairness. It involves the distribution of the quality of life and well-being of contemporary people, and ensures the fair provision of benefits to local communities. Contemporary people should recognize and strive to make their own opportunities equal to those of future generations while creating and pursuing the development and consumption of the present.

(3) Pay attention to the relationship among factors. It aims to combine the economic, environmental and social sectors in a balanced way to solve conflicts. Many studies are devoted to describing the complex interaction between and within sustainable development goals of social, economic and environmental dimensions of sustainability. [8].

The coastal and marine environment provide a wide range of direct and indirect benefits to the community, including food sources, health benefits such as increasing physical activity, education, continuation of indigenous culture and customs, and support for coastal industries such as fishing [10]. The United Nations Conference on Environment and Development clearly pointed out in Chapter 17 of Agenda 21 that

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https://doi.org/10.1016/j.marpol.2023.105689
Received 31 October 2021; Received in revised form 18 May 2023; Accepted 25 May 2023
Available online 30 May 2023

oceans, coasts and marine resources need to be properly managed in the future [6]. In 2015, the United Nations General Assembly adopted 17 sustainable development goals. The purpose of these goals is to set achievable targets, which can be achieved as the 2030 sustainable development agenda. Goal 14: life below water (the "Ocean Goal") aims to " conserve and sustainably use the oceans, seas and marine resources for sustainable development". The "Ocean goal" includes those related to the conservation and sustainable management of marine and coastal ecosystems, impacts of ocean acidification, sustainable use of resources and capacity building. These targets aim to ensure that human beings benefit from the diversity, richness and biomass of marine species and the diverse interactions between these organisms and the marine environment. As the world's oceans are important providers of ecosystem services for human social and economic well-being, the conservation and sustainable utilization of marine resources have important interactions with poverty reduction (SDG1), food security and nutrition (SDG2), economic growth (SDG8), urban development (SDG11), sustainable consumption and production (SDG12) and climate action (SDG13), especially for small island developing states [26].

As a strong comprehensive and cross-cutting research field, sustainable development assessment involves many disciplines and can be carried out with different emphases. In the past two decades, the connotation, assessment methods, assessment indicators, data sources and monitoring methods of marine sustainable development have been paid more and more attention, and many indicators and methods have been developed and applied to global and regional sustainability assessment [6].

The main purpose of this study is to put forward a general outline of marine sustainable development assessment through systematic review. We expound this outline from three aspects. The first goal is to analyze the connotation and conceptual framework of marine sustainable development through systematic literature analysis. The second goal is to establish a system to show the framework of marine sustainable development assessment, including assessment objectives, assessment methods, assessment scope, data collection, assessment results analysis and discussion of policy recommendations. The third goal is to analyze the limitations of marine sustainable development assessment and propose future research areas. By achieving these goals, we intend to make a clear impression of this gradual process, paving the way for future research on marine sustainable development assessment.

The ensuing section of this article provides methodology, focusing on explaining the process of system review.The third section is the description of the literature, and the findings are placed in the fourth section. The last section of the article shows the main conclusions and suggestions.

2. Methodology

Compared with traditional review, systematic review has the advantages of reliability, transparency, rigor and repeatability [13–15]. Therefore, this study adopts a systematic review approach to ensure effective and accurate evaluation of relevant literature.

The research group organized two reviews. The first review was conducted on August 18th, 2021. Taking marine sustainable development assessment as the key word, we have conducted extensive searches in several online databases (such as ISI, Web of Science, Emerald, EBSCOhost, Springer, Wiley and Proquest), and there was no time limit for the articles to be reviewed. The second review was on February 12, 2023. With marine sustainable development assessment as the key word, we consulted Scopus-Elsevier database, and the deadline for articles to be reviewed is 2022. Fig. 1 explains the step-by-step method of the review conducted here.

We got a total of 1547 articles through two database searches and after eliminating duplicate papers and non-journal articles, there were 1017 articles left. For 1017 articles, we conducted two rounds of in-depth research, of which 569 articles were excluded by the abstract in-depth research survey and 389 articles were excluded by the full-text in-depth research survey.In order to avoid negligence in the review, the review team re-examined 958 excluded articles. Finally, 59 papers on marine sustainable development assessment were screened out. Systematic review of these papers, in-depth analysis of the concept of marine sustainable development, and clear objectives for marine sustainable development assessment. The assessment process is the core of marine sustainable development assessment, including the selection of assessment methods, the determination of assessment scope and the acquisition of data. The evaluation results pay more attention to finding the current problems and analyzing the problems, and on this basis, put forward policy recommendations and suggestions for future research.

3. Description of literature

3.1. Publication activity

Articles on marine sustainable development assessment have been published in 36 kinds of marine management journals and journals related to environment, ecosystem and sustainable development research , as shown in Table 1. As can be seen from the table, the two journals that publish the most articles on marine sustainable development assessment are Ocean and Coastal Management and Plos One.

Judging from the publication time of the article, the research on marine sustainable development assessment was not paid attention until 2000. It can be seen from Fig. 2 that before 2000, there were only 2 research articles on marine sustainable development assessment, and in the next 12 years, 14 articles paid attention to this problem, and the number of articles increased significantly after 2012. It should be said that with the warming of climate and the gradual appearance of marine problems, researchers' attention to marine sustainable development has reached a peak.

3.2. Literature overview

Analysis was conducted on 59 research papers that focused exclusively on marine sustainable development assessment. The research on marine sustainable development assessment shows obvious stage characteristics. Through analysis, we can conclude the following three stages.

The embryonic stage (Before 2000). In 1992, the Marine Declaration of UNCED proposed: (1) to prevent, reduce and control the degradation of the marine environment in order to maintain and improve its ability to sustain life and production; (2) to develop and increase the potential of marine living resources to meet human nutrition needs and social, economic and development goals; (3) to promote integrated management and sustainable development. The Conference on Environment and Development promoted researchers' thinking on the sustainable development of the ocean. This stage focuses on sustainable development of large marine ecosystems and discussion of integrated coastal zone management [16,17]. The definition of marine sustainable development revolves around the changing ecosystem state and "health" index, focusing on the resilience, stability, productivity, biodiversity and yield of the changing ecosystem state[16].

The developing stage (2000–2012). The World Summit on Sustainable Development (also known as Rio +10), held in Johannesburg in 2002, proposed to assess the progress of oceans and coasts in the decade since the Rio Conference and identify new and persistent challenges [18]. At this stage, researchers realized that the study of marine sustainable development must go beyond the disciplinary boundary between natural science and social science, and must be rooted in the concept of ecosystem sustainability [18], and "society" should be included in the assessment of ecological sustainable development [5] to support the comprehensive management of our planet. The assessment methods of marine sustainable development also showed a characteristic of diversification in this period, and the DPSIR analysis method [19,

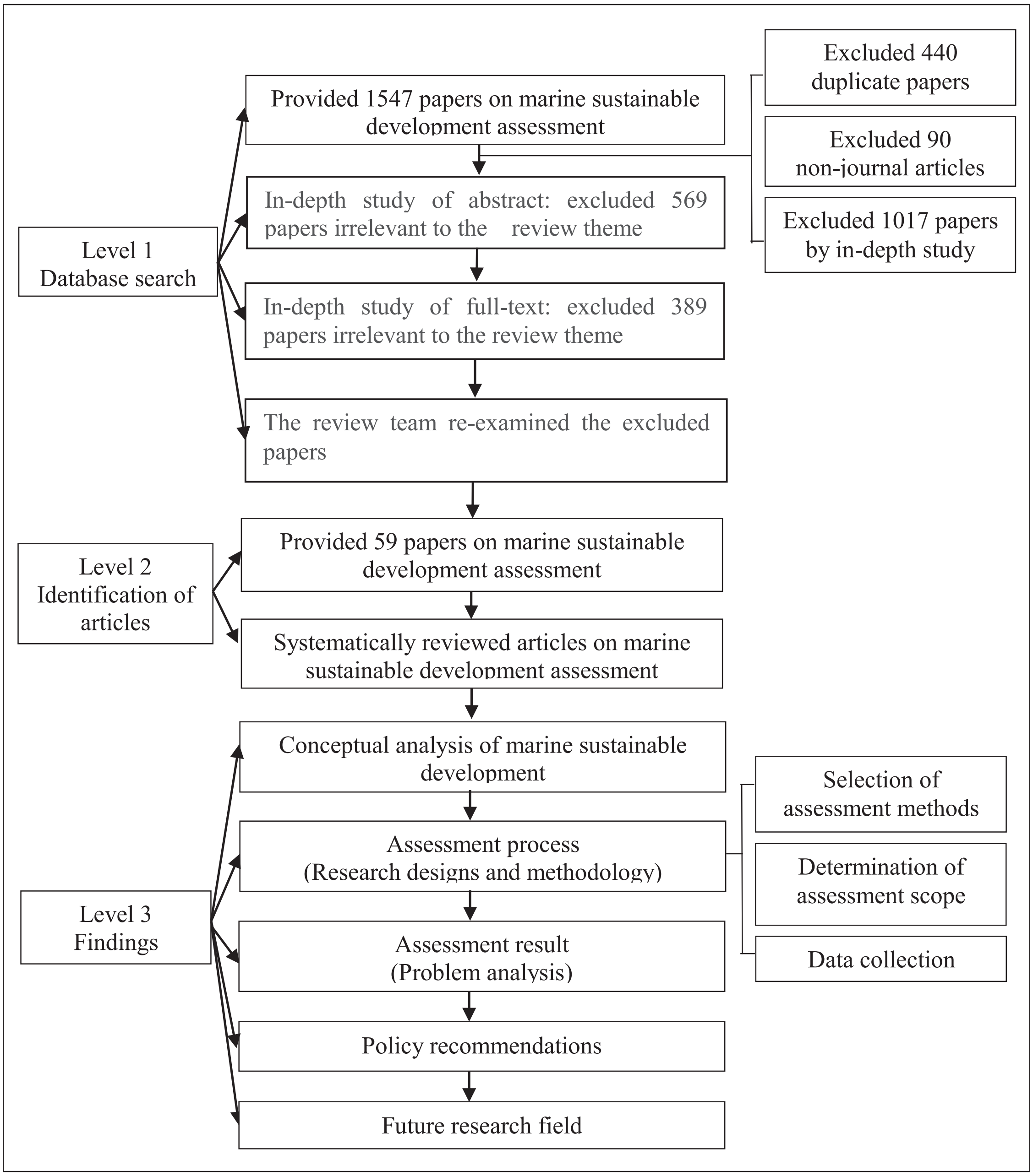

Fig. 1. Analysis framework.

20], energy synthesis method [21] and system dynamics model [22] were used to evaluate marine sustainable development.

The mature stage (After 2012). The outcome document of the 2012 United Nations Conference on Sustainable Development (or Rio+20) called for an inclusive and transparent intergovernmental process to formulate a set of action-oriented and universally applicable sustainable development goals [8]. In 2015, the United Nations officially set 169 goals to measure the progress of achieving sustainability under the 17 sustainable development goals. Among them, goal 14: life below water, which recognizes the importance of the oceans to human livelihoods and seeks to promote the conservation and sustainable development of oceans and marine resources [6,11,12,22–27]. The World Ocean Assessment is widely regarded as the main tool to guide the actions of SDG14. Under the heavy pressure of consumption, the natural environment of human beings is still deteriorating, and the role of science in dealing with climate change and biodiversity loss is highlighted. During this period, there were fewer diversified discussions about marine sustainable development, and more researchers discussed the formulation of the target framework of marine sustainable development around SDG14.

4. Findings

4.1. Concept analysis

More and more efforts are devoted to describing the sustainability at different levels[8]. Dividing sustainability into weak, strong and super-strong is helpful to better understand the relationship between different levels. The position of weak sustainability emphasizes the possibility of capital substitution and the power of technological process to alleviate resource depletion and pollution [4], which is closely related to the 2030 agenda. Most of the targets of Sustainable Development Goal 14 cannot be measured quantitatively, especially at the global level [11]. Goals are presented independently, but in social ecosystems, these

Table 1
List of basic information of literature.

Dimension model	Number of articles	Assessment methods	Number in references	Name of journals
Two-dimensional model	13	Compatibility evaluation of aquaculture and MPA objectives	[25]	AQUATIC CONSERVATION: MARINE AND FRESHWATER ECOSYSTEMS
		Energy Synthesis method	[21]	ECOLOGICAL QUESTIONS
		Interdisciplinary comprehensive evaluation method	[65]	ECOLOGICAL RESEARCH
		Interdisciplinary comprehensive evaluation method	[63]	ECOSPHERE
		Comprehensive evaluation method of ecological environment	[58]	FRONTIERS IN MARINE SCIENCE
		Interdisciplinary comprehensive evaluation method	[45]	ICES JOURNAL OF MARINE SCIENCE
		Driving force-pressure-state analysis	[60]	MARINE POLLUTION BULLETIN
		Index of health and benefits of the global ocean	[66]	NATURE
		Assessment method is not discussed.	[17]	OCEAN & COASTAL MANAGEMENT
		Driving force - pressure - state- response - control" (D-PSR-C)method	[35]	
		Qualitative method (Seminar)	[7]	PEOPLE AND NATURE
		Threat assessment	[57]	PLOS ONE
		Integrated risk-based assessment	[64]	SCIENCE OF THE TOTAL ENVIRONMENT
Three-dimensional model	13	Fishery Performance Indicators(FPIs)	[31]	PLOS ONE
		Review and evaluate sustainable management	[23]	INTERNATIONAL ENVIRONMENTAL AGREEMENTS-POLITICS LAW AND ECONOMICS
		System Dynamic method	[42]	JOURNAL OF COASTAL RESEARCH
		Qualitative method (Questionnaire survey)	[5]	MARINE POLICY
		Assessment method is not discussed.	[8]	
		DPSIR method	[11]	
		Qualitative method	[10]	OCEAN & COASTAL MANAGEMENT
		Literature analysis method	[28]	
		OHI method	[43]	PEOPLE AND NATURE
		Review of econometric models	[29]	REVIEWS IN AQUACULTURE
		Qualitative method (Scoring method)	[6]	SUSTAINABILITY
		Analysis on the rationality of sustainable development index	[27]	
		System Dynamic method	[30]	SUSTAINABLE COMPUTING-INFORMATICS & SYSTEMS
Four-dimensional model	3	System Dynamic method	[22]	ECOLOGICAL MODELLING
		Assessment method is not discussed.	[32]	SUSTAINABLE DEVELOPMENT
		Fuzzy group multi-criteria decision making approach	[33]	TRANSPORTATION RESEARCH PART D-TRANSPORT AND ENVIRONMENT
Five-dimensional model	1	Multi-criteria decision method	[34]	ENVIRONMENT DEVELOPMENT AND SUSTAINABILITY
The dimension model is not discussed.	29	DPSIR method	[38]	ECOLOGICAL ECONOMICS
		Process for assessing the state of the marine environment	[40]	AUSTRALIAN JOURNAL OF MARITIME AND OCEAN AFFAIRS
		Biodiversity assessment method	[24]	BIODIVERSITY INFORMATION SCIENCE AND STANDARD
		Large Marine Ecosystems assessment method	[49]	DEEP-SEA RESEARCH PART II-TOPICAL STUDIES IN OCEANOGRAPHY
		Evaluation method of marine protected areas	[55]	ECOLOGICAL APPLICATIONS
		Large Marine Ecosystems assessment method	[50]	ENVIRONMENTAL DEVELOPMENT
		Qualitative method (Seminar)	[61]	ENVIRONMENTAL TOXICOLOGY AND CHEMISTRY
		Qualitative method (Seminar)	[62]	
		Review and evaluate sustainable management	[18]	FRONTIERS IN ECOLOGY AND THE ENVIRONMENT
		Analysis of the role of world ocean assessment	[47]	FRONTIERS IN ENVIRONMENTAL SCIENCE
		Review of world ocean assessment.	[41]	FRONTIERS IN MARINE SCIENCE
		Investigation on the understanding of world ocean assessment	[46]	
		Environmental objectives indicators analysis method	[51]	GEOGRAPHY AND NATURAL RESOURCES
		Ecosystem health index	[16]	MARINE ECOLOGY PROGRESS SERIES
		Qualitative method (Seminar)	[12]	MARINE POLICY
		OHI method	[44]	
		Review of sustainable management policies	[52]	
		Evaluation method of marine ecosystem service value	[53]	
		DPSIR method	[37]	OCEAN & COASTAL MANAGEMENT
		Holistic model of fishery sustainability indicator	[39]	

(continued on next page)

Table 1 (*continued*)

Dimension model	Number of articles	Assessment methods	Number in references	Name of journals
		Review of Large Marine Ecosystem assessment	[48]	
		Evaluation method of marine protected areas	[56]	
		Qualitative method (Questionnaire survey)	[59]	
		Heuristic assessment framework	[9]	PLOS ONE
		DPSIR method	[36]	PROCEEDINGS OF THE 13TH INTERNATIONAL CONFERENCE ON ENVIRONMENTAL SCIENCE AND TECHNOLOGY
		DPSIR method	[19]	REGIONAL ENVIRONMENTAL CHANGE
		DPSIR method	[20]	
		PIR method	[26]	SUSTAINABILITY
		Multidimensional evaluation method	[54]	

Note: The articles listed in this table are 59 articles obtained by systematic review, and the codes in the references are 5–12 and 16–66, which have been listed in the table.

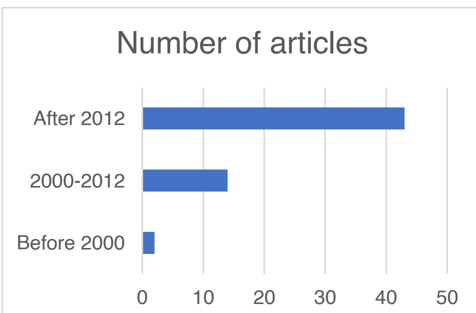

Fig. 2. Differences in the number of articles published at different times.

goals are often interrelated and interdependent, which means that the progress of one goal can promote or influence a series of other goals [12]. As the main evaluator of the progress of sustainable development goals, the evaluation based on indicators has become a common practice in all scales [26]. However, there are also many difficulties in the evaluation of indicators, such as the inconsistency of indicators and the lack of official data [11].

According to researchers' different understandings of sustainable development levels and interrelationships, our conceptual analysis of marine sustainable development is as follows.

(1) Two-dimensional model. The increase of population and the intensification of human activities are putting more and more pressure on the environment. Sustainability never comes to an end, because the balance between development and environmental protection must be constantly readjusted [17]. The core of the problem is not only to take plans and actions to rebuild the natural balance, but also to promote balanced economic growth and social and economic development of environmental protection in order to achieve sustainable development [19]. In coastal areas, decisions on the use, development and protection of natural resources must be balanced between public and private sectors, between environmental laws and ecological feasibility, between local and regional development, between national and international requirements, and between short-term productivity and long-term sustainable development [17].

(2) Three-dimensional model. The three pillars of sustainability, namely, the environment, society and economy promoted during the Rio Summit. Only by balancing these three levels in a comprehensive and holistic way can we achieve sustainability [5,9,10,25–29]. The marine sustainable economic development system is a marine ecological-economic-social complex system, including the sustainable transformation of population, intensive management and sustainable utilization of marine resources, virtuous circle of marine ecological environment, optimization of marine economic structure and industrial structure, and sustainability of marine economy [30]. This kind of balance introduces the idea that where there is no synergy between human beings and nature, it is necessary to weigh the three to achieve a sustainable balance. Another three-dimensional evaluation model, namely the fishery performance index (FPIs) method, is to evaluate fishery performance according to ecological, economic and community outcomes, and is considered as a widely applicable and flexible tool [31].

(3) Four-dimensional model. Sherman [16] proposed that the four subsystems of social economy, environment, ecology and management should be combined with the sustainable development model to comprehensively evaluate specific problems. Sustainable development model configuration is further divided into four subsystems-social economy, environment, ecology and management. Coelho et al. [32] proposed four levels, namely, environment, society, economy and system. Although there are differences in terms between them, in fact, there is little difference in connotation, and both of them add the dimension of human intervention to the three-dimensional theory. Ren & Liang [33] proposed that the four dimensions of sustainability are environment, economy, society and technology, among which technology as another dimension of sustainable development has attracted people's attention. The management, system and technology mentioned by the researchers all need innovation to promote marine sustainable development. In this case, the four dimensions can also be upgraded to society, economy, environment and innovation.

(4) Five-dimensional model. Ghorbanian & Zibaei [34] proposed sustainability dimensions which covered ecological level, economic level, social level, technical level and management level. In the human sustainable development system, economic sustainability is the foundation, ecological sustainability is the condition, social sustainability is the purpose, technological sustainability is the driving force, and management sustainability is the guarantee. The five parts complement each other and are inseparable. What human beings should pursue is the sustainable, stable, healthy and coordinated development of the complex giant system of ecology-economy-society-technology-management. Culture, as an important support for economic and social development, has been paid little attention to by relevant research so far.

Summarizing the above viewpoints, we find that sustainable development is a new paradigm, and its essence is to find a balance point at different development levels through constant adjustment, as shown in Fig. 3. Seeking a balance between development and environmental protection, people put forward a two-dimensional idea of sustainable development. Environmental protection, as a major focus of sustainable development, has never changed. People put forward new views on another dimension-development, which is subdivided into economic development and social development. Therefore, the formulation of three pillars is widely accepted, that is, the balance of environment (or ecology), society and economy can achieve sustainability. By adding the dimension of human intervention in the three-dimensional theory,

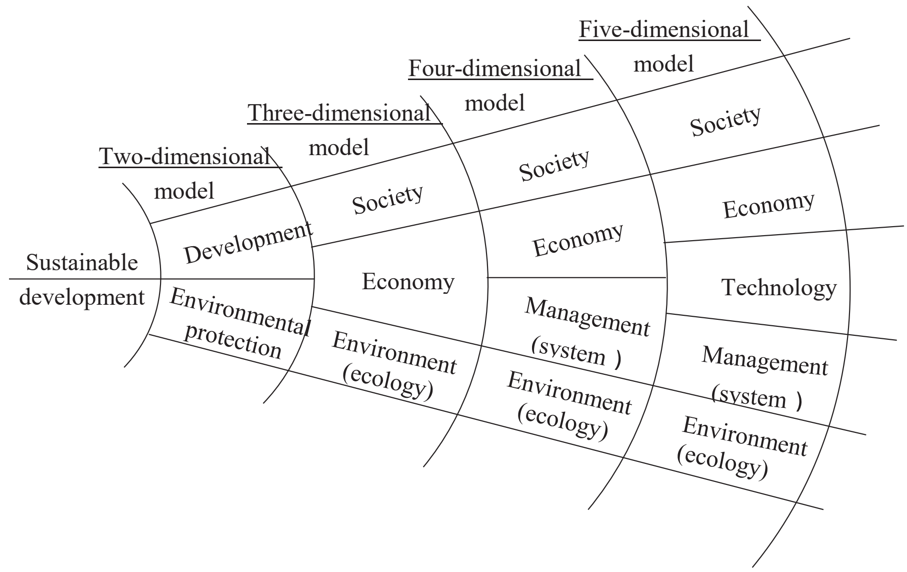

Fig. 3. Conceptual framework.

management has been promoted to the same level as environment (or ecology), society and economy, that is, a four-dimensional model has been put forward, which shows people's deeper thinking on sustainable development. The importance of technological progress for human intervention gradually appears, and the five-dimensional viewpoint of environment (or ecology)-economy-society-management-technology is more conducive to promoting the sustained, stable, healthy and coordinated development of the complex giant system.

4.2. Assessment process

4.2.1. Selection of assessment methods

There are various methods for researchers to evaluate marine sustainable development, including quantitative evaluation and qualitative analysis. Here we list the common evaluation methods, and other evaluation methods are shown in Table 1.

4.2.1.1. DPSIR method. The Pressure-State-Response (PSR) model originated in 1970 s, which was proposed by two Canadian scholars, and was developed by the Organization for Economic Cooperation and Development (OECD) in 1993 to form a PSR framework for environmental performance evaluation. The PSR model is a method to construct an evaluation index system from the mechanism of index generation. The PSR model mainly includes three kinds of indicators [32]: pressure, state and response, which are mainly used to reflect the interaction between human beings and the natural environment.

The model of "driving force-pressure-state-impact-response" (DPSIR) was put forward by the European Environment Agency (EEA) after improving the model of "Pressure-State-Response" (PSR) in 1998. The model includes social, economic and environmental factors. It expresses the interaction between people and the environment from a systematic point of view, and can better reflect the relationships among the various elements of the complex ecosystem of tourist destinations. The DPSIR model describes the dynamics of the system under consideration [33], which is consistent with the ecosystem approach in marine environment [11]. Its operating mechanism is that the development of social economy and population acts as a long-term driving force (D), thus exerting pressure on the environment [34], causing changes in the state (S) of marine ecological environment, thus causing various impacts (I) on the marine ecological environment and socio-economic environment. These impacts prompt human beings to respond (R) to the changes in the state of marine ecological environment [18,24].

With the wide application of sustainable development theory, more quantitative research methods are needed for the evaluation of sustainable development. Therefore, the evaluation model contains more indicators, especially in the marine field. The DPISR model is widely favored by researchers in marine sustainable development assessment [11,18,24,30,32–35,39], including the First World Ocean Assessment and the Second World Ocean Assessment [35,36]. The advantage of the DPSIR model is that it can solve complex problems involving various factors in multiple dimensions. The DPISR model can build a dynamic framework, which can not only integrate indicators, but also explain the dynamic mechanism of marine sustainable development from different angles. Direct feedback of driving force, pressure, state or influence through actions [36] [37] will help decision-makers to take regulatory measures for sustainable development.

4.2.1.2. System Dynamic method. System dynamics is a powerful and simple method, which uses causal loop and stock flow chart to describe interrelated systems. A large number of literature shows that revealing the complex dynamic behavior of sustainable management system is an appropriate method. Using the concept of integrated coastal zone management, Chang et al. [22] developed a decision support system based on system dynamics. The four subsystems of society, environment, ecology and management are combined with the sustainable development model, and the specific problems are comprehensively evaluated. A sustainable development model is established as a decision support system to promote scenario analysis and solve complex coastal area management problems. Based on the study of offshore ecosystem, Yang & Wang [30] established a system dynamics model for the sustainable development of offshore economy. Through theoretical analysis, method discussion and case application, the external value assessment of marine ecosystem was comprehensively studied. Li & Li [42] used system dynamics and model analysis to qualitatively and quantitatively evaluate the sustainable development of marine economy. Generally speaking, the survey results show that people pay more and more attention to environmental quality, living environment and resource supply capacity, which is in line with the actual situation.

4.2.1.3. OHI method. Ocean health index (OHI) is a comprehensive index to evaluate the ability of the ocean to provide human welfare and its sustainability. OHI quantifies and scores a series of social and ecological benefits and ecosystem services provided by the ocean for human beings (called goals in OHI) [40], so as to reveal the health status and changing trend of the ocean. Based on the OHI method, a large number of existing social ecological data were used to assess the health status of nine targets in the Baltic Sea [40]. The semi-enclosed Baltic Sea is a typical example of a brackish water ecosystem affected by various artificial pressures, such as eutrophication, rising levels of harmful substances, introduction of non-native species and habitat degradation, and unsustainable fishing pressure. The analysis shows that the indexes in OHI database provide better insights on data structure and usability [41]. OHI evaluation uses open data to ensure that the method is transparent and repeatable. It is considered as a valuable comprehensive evaluation tool, which can provide information for Ecosystem-based Management (EBM) and track the progress of achieving sustainable development goals [40].

4.2.1.4. Effectiveness of marine protected areas. In coastal and marine management, marine protected areas are more and more used to promote sustainable development, not just to protect [56]. Most of the areas that need protection are inhabited and used by human beings, which means that the ecological protection goal must be balanced with the social and economic background. At present, the assessment of the effectiveness of marine protected areas can be divided into quantitative and qualitative assessments. The qualitative evaluation of the effectiveness of marine protected areas is mainly based on semi-structured interviews, which combine the protection objectives with the user 'perspective and involving stakeholders in planning and management. Morf et al. [56] adopted a comparative case study design to explore the interplay between the institutional arrangements and management outcomes in two adjacent yet institutionally slightly differing marine protected areas. The quantitative evaluation method is mainly used to calculate the protective effect of fish in marine protected areas. For example, researchers have developed an effective method to assess the allocation of marine protected areas, which is used to assess the sustainability and yield of populations with sedentary adults and dispersing larvae [55].

4.2.1.5. Evaluation of marine ecosystem service value. Marine and coastal ecosystems provide myriad of services to humans—a rich variety of seafood, regulating climate and water resources, and protecting human beings from storms and tornadoes. The assessment of marine ecosystem services can help inform decision-making, and these services are essential to support and strengthen the livelihoods and well-being of indigenous and local communities across the globe. The framework of ecosystem services value assessment uses the Common International Classification of Ecosystem services (CICES) framework [53]. According to the CICES framework, three categories of ecological services are used to evaluate their important roles in the local economy and people's well-being: supply, management and maintenance, and culture. Specific evaluation methods include market price method and indirect (linked) market method. Market price method refers to direct market pricing, expenditure, replacement cost and avoidance cost, and indirect (linked) market is used to infer value, that is, hedonic pricing and travel cost. Sangha et al.[56] evaluated the value of coastal and marine ecosystem services statewide, and provided information for the sustainable development policy of the Northern Territory of Australia.

4.2.2. Determination of assessment scope

In order to achieve a comprehensive assessment of marine sustainable development, transboundary and land-marine spatial factors should be considered [42], and the main intervention spatial scale determines the most relevant geographical span for formulating indicators: global, regional, national and local [11,40].

4.2.2.1. Global assessment. Two World Ocean Assessment reports are the most famous for the global scale marine sustainable development assessment. In 2002, the United Nations General Assembly approved the establishment of a regular process for global reporting and assessment of the state of the marine environment, including socio-economic aspects. The first cycle of the regular process was conducted in the period of 2010–2015, and the first global integrated marine assessment was completed in the second cycle, which was widely regarded as the main tool to guide the actions of Goal 14 of Sustainable Development [43]. This assessment is the first comprehensive global overview of the state of the ocean and the relationship between the ocean and human beings, covering environmental, social and economic aspects [44]. The first World Ocean Assessment report pointed out that some marine environments, especially those near the coast, are seriously degraded and the marine conditions are generally declining. Therefore, comprehensive, coordinated, proactive, cross-sectoral and science-based methods should be adopted for coastal and marine management, so as to avoid the continuous decline of the ocean's ability to provide important services to society and the whole planet[44].

Since the first assessment provided baselines for many aspects of marine socio-economic and biogeochemical systems, a key focus of the second World Ocean Assessment was to assess possible changes since the first World Ocean Assessment based on these baselines [38]. The second assessment pointed out that due to the increasing pressure exerted by human beings on the ocean, many aspects of the ocean continued to decline, and further highlighted some obstacles existing in the implementation of these solutions, including major restrictions on resource capacity (including financial capacity) and technical capacity[44].

4.2.2.2. Regional assessment. Large marine ecosystems are regions of marine space, which are relatively large areas of about 200,000 square kilometers or more, and are characterized by unique bathymetry, hydrogeology, productivity and nutrient-dependent populations [48]. Their ecologically defined boundaries generally include the coastal marine areas of two or more countries [44,45]. They serve as global centers of marine socio-economic development, including marine sectors that maintain fisheries, energy, transportation, recreation and mining. Large marine ecosystem assessment is multidisciplinary and multisectoral, as a part of the global movement to maintain the world's large marine ecosystems [50]. After UNCED, the project of transnational cooperation to study, monitor and evaluate the health status of large marine ecosystems received attention and support [16]. Shul'kin et al. [51] discussed the list of environmental targets and indicators for monitoring the state of marine environment in the Northwest Pacific region. The sustainable development of large marine ecosystems in Latin America and the Caribbean has attracted much attention, and with the assistance of the Global Environment Facility, Latin America and the Caribbean is carrying out large marine ecosystem assessment [47]. There are 27 small island developing States in the Caribbean Sea, some of which make use of the marine resources of the Caribbean Sea to obtain economic benefits, such as developing tourism, shipping and industry, etc [52]. The sewage, agricultural wastewater and hydrocarbon pollution resulted in the degradation of this large marine ecosystem [49]. The Baltic Sea is one of the fastest warming large marine ecosystems in the world. The Baltic Sea Health Index (BHI) is introduced as a transparent, collaborative and repeatable assessment tool. The results show that the overall health status of the Baltic Sea is not ideal, and the BHI scores of carbon storage, pollutants and permanent special places (i.e. marine protected areas) are the lowest [40].

4.2.2.3. National assessment. The selection of indicators for National Sustainable Development Goal 14 is based on existing global and regional indicators, supplemented by national indicators related to national policies or national stakeholders when necessary [11]. The French National Bureau of Statistics shared the first database containing 110 indicators of sustainable development goals in March 2017. The French Ministry of Ecology, Energy and Ocean is responsible for the complete set of indicators for Sustainable Development Goal 14 [11]. On the basis of summarizing the current comprehensive modeling of marine system in Australia, Melbourne-Thomas et al. [45] put forward the future needs of comprehensive modeling of marine social ecosystem in Australia by comparing and integrating the models, and provided a set of suggestions for formulating the priority areas of national comprehensive modeling methods. Hora [26] used the Sustainable Development Goals as the framework of comprehensive assessment, and selected indicators according to the scientific evidence of the interrelation between the Sustainable Development Goals, with the aim of developing a monitoring tool to help assess the national baseline for comprehensive implementation of the Sustainable Development Goals, and evaluated island countries. Gulseve [44] provided an indicator-based assessment guide for monitoring the progress of achieving Sustainable Development Goal 14, and assessed the achievements of the United Arab Emirates in achieving Sustainable Development Goal 14.

4.2.2.4. Local assessment. As a local area under the national category, it is concerned by researchers because of its small scope and easy access to indicator data. Based on the study of offshore ecosystem, Yang & Wang [30] established a system dynamics model for the sustainable development of offshore economy, evaluated China's coastal provinces and cities, and calculated the comprehensive development of innovative ecosystem of high-tech enterprises in Shanghai, Liaoning, Tianjin and Heilongjiang from 2014 to 2018. Sangha et al. [53] assessed the value of coastal and marine resources in the Northern Territory of Australia, and provided a reliable tool for ecosystem-economic assessment to support the fair and sustainable use of resources. Taking Nantong as an example, Wei et al. [35] calculated the comprehensive carrying capacity within a given space and time frame to support the evaluation of long-term sustainable development of the regional economy. Coelho et al. [32] designed a framework for formulating regional sustainability indicators, which proposed a set of common local indicators and involved local communities in regional sustainability assessment. Cerreta et al. [54] dealt with the conflicts in the overlapping areas of cities and ports through the principles and tools of multidimensional assessment, and constructed a four-step methodological framework, which was expressed and evaluated by a set of core sustainability indicators related to sustainable development goals.

The management of marine protected areas is a topic of interest to the scientific community, because it is necessary to maintain a balance between sustainable development and environmental protection in these areas [33]. According to the analysis of the selected indicators and application of DPSIR model, Monemvasioti & Tsoukala [36] made a comprehensive evaluation of the environmental conditions of Zakynthos Marine National Park. Franzese et al. [21] combined geographic information system and energy synthesis method to model the interaction among economy, environment and resources in marine protected areas, and took southern Italy as an example for empirical research. Kaplan et al. [55] developed an effective method to evaluate the allocation of marine protected areas, so as to evaluate the persistence and yield of a population with sedentary adults and dispersing larvae. Morf et al. [56] used a comparative case study design to explore the interaction between institutional arrangements and management achievements of two adjacent marine protected areas with slightly different systems, and analyzed the sustainability balance of two pioneering marine national parks. Guarnieri et al. [57] carefully evaluated the potential impacts of different pressure combinations (land and ocean pressures) on vulnerable rocky habitats along the 40 km coast of the western Mediterranean (Ionian Sea). In order to achieve the Aichi target of 10%, countries will need to greatly expand the designated scope of marine protected areas. According to current experience, most of them will belong to the category of multi-purpose marine protected areas [25].

4.2.3. Data collection

The assessment of marine sustainable development needs sufficient data support. (1) Ocean observation data is an important source of marine sustainable development assessment data. In the past decades, extensive work has been carried out under the formal framework of ocean observation at local, regional and global levels to expand and pay better attention to the continuous observation of the coastal and marine environment. Many of these gaps and limitations were elaborated in the First World Ocean Assessment, and these gaps and limitations were also confirmed in other assessments at local, regional and global levels [38]. Most observation networks did not extend to the economic, social and cultural aspects of the ocean. (2) Targeted survey data is also one of the important sources. Researchers select sites in the study area and obtain data by sampling[55]. Wei et al. [35] used the field survey data as one of the data sources, and further improved the index system to obtain research conclusions. (3) Some studies need to collect data and information through questionnaires and interviews. For example, Fawkes & Cummins [46] conducted such interviews, including the survey responses of 96 individuals from 35 different countries, and a series of 17 semi-structured interviews with marine and sustainable development experts from 10 different countries. Satumanatpan et al. [58] conducted an assessment of marine and coastal resource governance in Koh Tao, Thailand through a questionnaire survey. (4) Another data source is open source data, including websites, bulletins, statistical yearbooks and so on. As a source of sustainable development goals, the United Nations Statistics Division is currently developing an e-book on indicators and an indicator portal database. Another source of information is the Sustainable Development Goals Tracking Website [44]. Wei et al. [35] has collected statistical yearbooks, statistical bulletins, marine bulletins and historical data of Jiangsu Province and Nantong City for his research on the coastal areas of Nantong.

From the perspective of dimension model systems, different assessment dimensions need different data support: (1) the data needed for ecological or environmental dimensions mainly come from ocean observation data and field survey data; (2) the data needed for the economic dimension mainly comes from open source data released by government departments such as statistical yearbooks and statistical bulletins of various countries; (3) the data needed for the social dimension includes not only open source data released by government departments such as statistical yearbooks and statistical bulletins from various countries, but also data information collected through questionnaires and interviews; (4) the data needed for the technical dimension, such as some scientific research achievements, can come from some professional databases; (5) the data needed for the management dimension can come from open source data of government departments, and can also be collected through questionnaires and interviews.

4.3. Assessment result

Marine environmental problems have become a global problem, and marine sustainable development is challenged. The results of the two global marine assessments show that the state of the ocean is generally declining, and marine problems should be paid attention to [47]. The Pacific marine ecosystem is increasingly affected by pressure factors such as pollution, overfishing, ocean acidification, coastal development and warming events, sea level rise and extreme weather frequency increase [60]. These human-driven pressures accumulate on different spatial and temporal scales, resulting in the continuous and widespread degradation of many marine ecosystems in the Pacific Islands region. Europe faces major challenges in risk assessment and management of chemicals and other stressors. This limits the ability of the region to contribute to the achievement of global sustainable development goals [61]. The pressures affecting the sustainable development of Asia's large marine ecosystems come from overfishing, pollution, overnutrition, habitat degradation, biodiversity loss and climate change, which are affecting the economies of neighboring countries [49]. Advances in environmental toxicology, environmental chemistry, biological monitoring and risk assessment methods are necessary to solve the adverse effects of environmental pressure on ecosystem services and biodiversity, because Asia is home to many biodiversity hotspots [62]. The lack of knowledge, appropriate tools and skilled human resources for ocean management remains a major constraint for the protection and conservation of the marine environment in many regions [47].

4.4. Policy recommendations

The research of marine sustainable development assessment has enabled us to discover the existing problems in the ocean and put forward corresponding countermeasures. (1) National and international institutions have formulated the legal framework to realize the vision of sustainable development, and carried out research related to management. Both the 2030 United Nations Agenda for Sustainable Development and the 2021–2030 United Nations Decade of Marine Science for Sustainable Development recognize that sustainability is an ideal and necessary way to ensure that the ocean can continue to provide services that society depends on [47]. The United Nations announced the Decade of Marine Science for Sustainable Development and the Decade of Ecosystem Restoration (both from 2021 to 2030) to support "reversing the cycle of declining marine health and convening the efforts of global marine stakeholders" [43]. (2) The management system based on ecosystem has been proposed to solve some problems inherent in traditional management [18]. It is considered as a promising method to track and manage the state of the ocean because it takes into account species, habitats, departments and user groups [63]. By identifying the main threats to healthy marine ecosystems, the most effective management measures to mitigate these threats are put forward [64]. (3) It is necessary to strengthen international scientific and technological cooperation and expand ocean observation capabilities. At present, there is a gap in ocean observations used to support assessment, and it is necessary to develop knowledge to fill ocean observations [41].

5. Discussion and conclusion

In this study, the conceptual framework and evaluation framework system of marine sustainable development are analyzed by systematic review method. We provide specific research fields from which researchers can fully understand the current trends and knowledge of marine sustainable development assessment.

This systematic and well-organized review has made the following contributions: for the first time, a systematic literature review in the field of marine sustainable development assessment has been carried out; it summarizes the past research and highlights the characteristics of each link in the process of marine sustainable development assessment; it also puts forward the limitations of contemporary research literature that need to be solved in the future, and suggests the future research fields.

5.1. Limitations

(1) The basic theory is not perfect, and the comprehensive evaluation theory and method of marine sustainable development are not mature. Because people don't have a deep understanding of the connotation and structure of sustainable development, and their research has not yet formed a relatively comprehensive and complete system, people blindly study the sustainable development capability without thorough analysis of its concept and structure. In order to improve the scientificity and accuracy of the research on sustainable development ability, it is necessary to study its connotation and structure.

(2) Lack of index data. A large amount of data is needed to comprehensively evaluate the sustainable development of the ocean. Although great progress has been made in establishing ocean observation networks and developing related capabilities, there are still major limitations in obtaining comprehensive and timely ocean information [41,66]. The lack of data limits our understanding of ocean processes and activities across multiple time and space scales.

(3) The evaluation criteria are quite different. There are great differences in marine sustainable development assessment methods, and different assessment methods have different settings for marine sustainability assessment, which leads to the lack of comparability of assessment results between different levels and different regions.

(4) Lack of long-term tracking. In many cases, the changes of marine ecosystem affected by natural conditions and human activities are not short-term and need long-term observation. Scholars have paid close attention to marine sustainable development for a short time, and lack comparative analysis of long time series.

5.2. Future research fields

There are two trends in the development of scientific research, one is to develop in depth in disciplines, and the other is to develop in the direction of "integrity" or "systematicness". The research on marine sustainable development also has this trend, that is, it is developing in depth in the direction of systematicness [26], comprehensiveness [5] and intersectionality [8]. It is mainly manifested in: paying more attention to the cross-integration of natural science and social science [65], the organic combination of qualitative research and quantitative research, especially the comprehensive integration of global change response and sustainable development. The research on assessment process and assessment method will be further deepened, and marine sustainable development assessment will become the frontier field of regional sustainable development research. The comprehensive application and technical support of modern information technology in this field will be paid more attention to. Here we list several fields for future research.

(1) Deeply study the conceptual framework of marine sustainable development. Marine sustainable development is a comprehensive concept, which has been interpreted from different dimensions (including two-dimensional, three-dimensional, four-dimensional and five-dimensional). Human beings' cognition of the ocean is deepening, and their cognition of the sustainable development of the ocean is constantly updated. The importance of this concept is reflected in the fact that it is the basis for formulating the goal of marine sustainable development, and it affects the construction of the evaluation index system of marine sustainable development. Therefore, it is necessary to further explore the conceptual framework of marine sustainable development.

(2) Develop innovative methods for data collection of marine sustainable development assessment. Data should be more reliable, more frequent, more cost-effective and more classified [11], so as to ensure the scientificity and effectiveness of marine sustainable development assessment. We expect to link multi-scale observation with science and policy. Although significant progress has been made in establishing ocean observation networks, developing related capabilities and improving modeling and reporting procedures, there are still fundamental gaps in observation and major limitations in obtaining comprehensive and timely ocean information [41].

(3) Establish criteria for judging marine sustainability. The marine sustainable development indicators of regional marine entities are different in concrete degree, reasons for selecting indicators, complexity and use of qualitative information. No matter what the goal is, a baseline should be established, and further research on the minimum safety standards and key and non-key indicators will be very necessary [41].

(4) Carry out the trend analysis of long time series. Our attention to the marine ecosystem and its sustainable development is still relatively short. Obviously, it is important to reassess human utilization of marine resources if climate change and human activities threaten sustainable development. To understand our ever-changing marine ecosystem, it is necessary to develop comprehensive science, pay close attention to ocean changes for a long time, and carry out long-time series trend

analysis to solve the synergy, interaction and nonlinearity we have seen.

In addition, it is helpful to discuss the differences between developed and developing countries in determining future work and effectively implementing sustainable development. In the future, we can further explore how developed countries and developing countries can cooperate to meet the challenges of sustainability and how to better integrate the needs and priorities of developing countries into the sustainable development agenda.

We know that the assessment of marine sustainable development is only a means, and the ultimate goal is to make people realize the problems and difficulties encountered in the process of marine sustainable development, so as to formulate more effective, targeted and innovative public policies [11] and promote the complex giant system of "ecology-economy-society-technology-management" to develop in a more sustainable direction. What we need more is that government departments take effective measures, rather than staying on paper or verbally.

CRediT authorship contribution statement

Fenggui Chen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft. **Yuhuan Jiang:** Formal analysis. **Zhenghua Liu:** Formal analysis. **Ruijuan Lin:** Formal analysis. **Wei Yang:** Formal analysis.

Data Availability

No data was used for the research described in the article.

Acknowledgements

The authors are grateful for support provided by Asian cooperation fund project named Marine Protected Areas Network in China-ASEAN Countries, Demonstration Study on Ecological Protection and Management of Typical Bay.

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